
Effect of 2PP Printing Parameters on Print Fidelity and Mechanical Behavior of Micro-Columns

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Abstract

3D printing at the nano-scale and micro-scale is becoming increasingly used for applications including surgery, circuitry, and robotics, among others [1]. Therefore, it is of great importance to understand how both the structures and materials produced by these fabrication processes mechanically behave [2-4]. The chosen printing parameters influence whether a print will be successful or fail according to the relevant metrics, thus the research herein consisted of varying multiple of the primary laser, speed, and slicing-related parameters, while keeping the photoresist (IP-S), the sample geometry, and the printing device (Nanoscribe Photonic Professional GT+) constant [1,3]. Hence, the goal of this project is to develop a guide for choosing an acceptable combination of settings for given mechanical design requirements. So far, 2PP-printed columns have been elastically compressed via a nanoindenter in a set of pilot experiments. Furthermore, a survey of commonly used slicing distances, hatching distances, scan speeds, and laser powers was conducted to determine feasible ranges for fabrication of structures without major defects. Elastic material properties are reported for pilot experiments demonstrating the effects of solid versus scaffold printing modes as well as development times, and bounds on parameter values yielding structures with defects are also listed [3,6].

Introduction

Our research group has extensive experience printing microneedles for surgical delivery and aspiration in the cochlea [7,14,15], and it is of great interest to fully understand the compressive mechanical behavior of the material resulting from their specific printing recipes to better model the 3D structures [3,6]. Moreover, the papers referenced delve into more detail on how to design microscale devices to be fabricated with a Nanoscribe instrument, particularly dual-lumen microneedles.

The Nanoscribe Photonic Professional GT+ is a high-resolution micro-/nanofabrication platform based on two-photon polymerization (2PP), also referred to as direct laser writing or multi-photon lithography [1]. In 2PP, a tightly focused ultrashort pulsed near IR laser is scanned through a photosensitive resin [1,3]. Polymerization occurs only where the local photon density is sufficiently high for simultaneous absorption of two photons, which confines curing to a small 3D voxel near the focus [1,3,4]. This nonlinear optical phenomenon enables sub-micron feature capability and high shape fidelity, while the surrounding resin remains uncured and acts as a support medium during writing [1].

However, despite evidence that parameters such as laser power, scan speed, and more impact the mechanical properties of 2PP-printed polymeric structures, the specific properties of our group's typical printing recipes remain relatively unstudied. Thus, the project to characterize the IP-S prints began. After completing the appropriate user training, an initial set of micro-column prints was conducted using a baseline set of parameters similar to those typically used for the group's microneedles [14,15]. Multiple iterations of these columns were printed while varying the interior geometry and development time to analyze the effects of these parameters on the elastic response of the structures.

Once a nanoindentation procedure was refined and baseline elastic response curves were obtained, understanding the limits of successful printing was necessary [3,4]. For this, several matrices of design parameters were prepared in the Software DeScribe. Iterative analysis of the resultant print geometries was then conducted to check for defects and prepare to experimentally measure the elastic and plastic behavior of the columns and observe how the response evolves as the parameters change [4-6].

Certification

In order to print the necessary columns for the entire project, the project members had to get certified to use the Nanoscribe at the CUNY Advanced Science and Research Center (ASRC) in upper Manhattan [1]. The certification required a virtual training session on chemical use and disposal, and emergency protocols. Once certified with the virtual training, an in-person training on the Clean Room facilities was scheduled and completed. Lastly, in-person training on the Nanoscribe printer itself, as well as its operating software NanoWrite, was completed [1].

Methods: 2PP with Nanoscribe Photonic Professional GT+

Printing files were developed using the parameter sweep workflow within the Advanced STL Processing menu of DeScribe slicing software. STL files were generated in Solidworks CAD software before import and columns were prescribed a diameter and height of 100 microns. Samples were then fabricated using Dip-in Laser Lithography mode with Nanoscribe's proprietary photoresist IP-S on a polished silicon wafer substrate. A laser power of 100% was chosen for all prints for the body of the structures. Prints were developed for 25 min (unless otherwise indicated) in SU-8 developer, also known as PGMEA, and two subsequent 10 min sessions in IPA.

Methods: Optical Microscopy

The fidelity of the sample geometry was visually analyzed using optical microscopy on a Keyence VHX 5000 microscope.

Methods: Compressive Elastic and Plastic Deformation Testing

Micro-column compression tests were conducted using a Keysight G200 nanoindenter by indenting with a stainless steel, flat, 1 mm punch to a set displacement of ~3 microns (approx. 3% strain) at a set strain rate of 0.05 1/s. To secure the samples for testing, glass microscope slides were adhered to sample holders using CrystalBond heat-activated adhesive. After cooling, the as-printed samples on the silicon substrate were secured to the mounted glass slide using 2-part epoxy to protect the structures from temperature-induced material changes.

An "effective Young's modulus" using engineering stress and strain with the assumption of uniaxial stress was computed using the unloading portion of the force-displacement curves obtained.

Results: Initial Printing and Testing (Spring 2025)

As a baseline, the initial columns had the following parameters, as based on the previous microneedle prints: laser power = 100%, base laser power = 60%, scan speed = 100000 microns/s, adaptive slicing distance [0.5 microns to 1 micron], and hatching distance of 0.5 microns [14,15]. It was unclear how the mechanical response of solid columns and columns with interior scaffolding differed, so initial files were made for both types of columns and then printed ("columns v3 solid" and "columns v3 scaffold") [4,6]. In total, nine sets of four columns were printed, totaling 36 columns [5,6]. From these tests, a subset of solid (N = 12) and scaffold columns (N = 11) that underwent development for 25 min in PGMEA and 20 min in IPA were elastically compressed. Additionally, an additional subset of solid (N = 8) and scaffold (N = 4) samples were developed overnight for approximately 1230 minutes in PGMEA and 20 min in IPA before elastic compression.

An effective Young's modulus of ~3.5 GPa (N = 12) was obtained for solid columns with a short development time and ~2.9 GPa (N = 11) for scaffold columns with a short development time, while solid columns and scaffold columns with a long development time had moduli of ~3.3 GPa (N = 4) and ~2.7 GPa (N = 4). Results from one set of solid samples with a long development time were discarded due to damage during the sample preparation process. Representative curves are shown in Figures 1 and 2 below.

In the future, tests may be repeated with larger aspect ratio columns to better emulate uniaxial stress conditions for estimation of the compressive elastic modulus. [6]. Alternatively, inverse Finite Element Analysis can be utilized to verify the elastic modulus results for the geometry tested herein [4,5].

Figure 1: Representative force-displacement curves obtained via nanoindentation for a batch of the solid columns with a short development time.

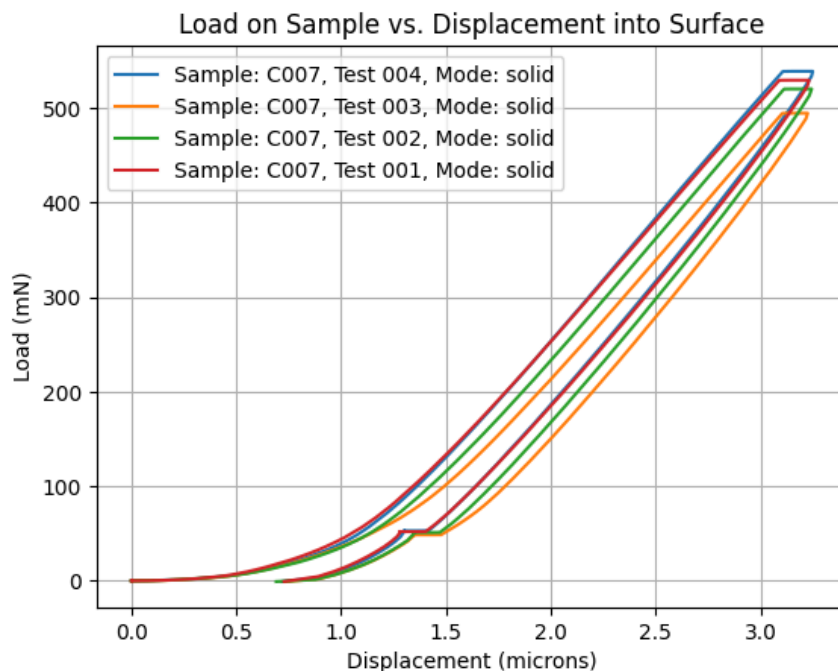
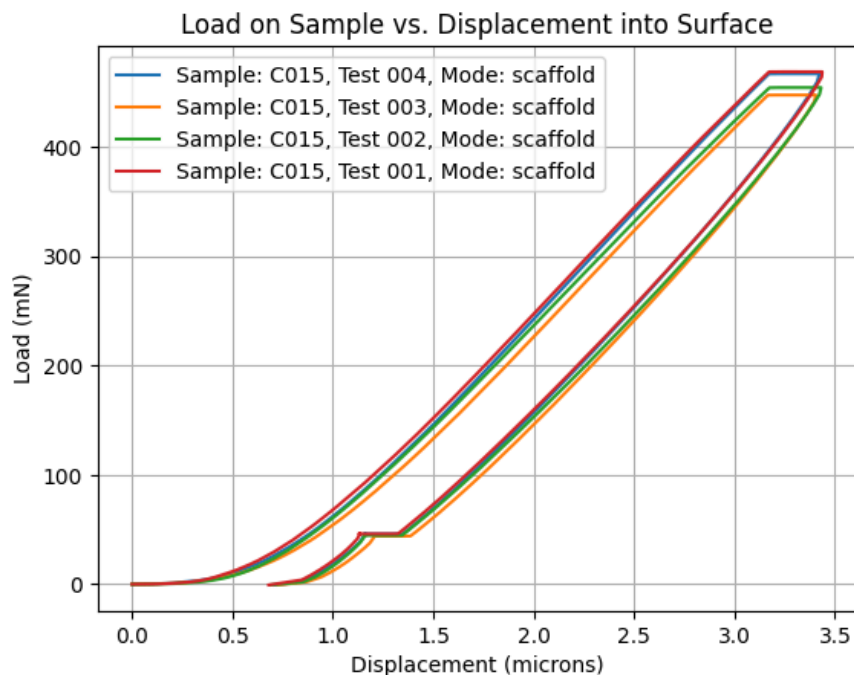


Figure 2: Representative force-displacement curves obtained via nanoindentation for a batch of the scaffold columns with a short development time.



Results: Parameter Sweep (Fall 2025)

Once the pilot experiments were completed, it was important to plan what parameters needed additional testing, and how many columns could be printed for said testing. The varying parameters selected were contour/hatching distance (lateral printing spacing), slicing distance (vertical printing spacing), and shell scan speed (velocity at which the laser travels).

The printing fidelity results are included in the tables below, with “X” indicating the presence of bubbling within the photoresist due to overexposure to the laser during printing and “✓” indicating a defect-free sample. First, a 3 by 6 matrix was designed in DeScribe (refer to Table 1). Next, a 5 by 2 parameter matrix was designed to test the effects of varying the base laser power values at the extreme slicing and contour distances to determine an acceptable value for the subsequent batches. After determining that the slicing distance likely had a larger effect on the creation of bubbles than overexposure due to reflection of the laser beam at the base of the structure, the slicing distance range was increased for the second 3 by 6 matrix. A final 3 by 6 parameter matrix was then constructed by increasing the slicing distances even higher, using the successful results from the base power sweep test as a lower bound. Additionally, figures demonstrating some of the observed defects are included.

Three sets of these 18-column batches were printed in preparation for elastic and plastic mechanical characterization with the nanoindenter (not yet completed).

Table 1: Columns v4 Sweep. Base laser power = 60%.

		Slicing Distance (μm)					
		0.25	0.25	0.5	0.5	0.75	0.75
contour distance (μm)	0.75	X	X	X	✓	✓	✓
	0.5	X	X	X	X	X	✓
	0.25	X	X	X	X	X	X
		75000	100000	75000	100000	75000	100000
		Scan Speed (μm/s)					

Table 2: Columns v4.1 Power Sweep.

		Slicing Distance (μm)	
		0.25	0.75
		Contour Distance (μm)	
		0.25	0.75
		Scan Speed (μm/s)	
		75000	100000
base laser power (%)	60	X	✓
	55	X	✓
	50	X	✓
	45	X	✓
	40	X	✓

Table 3: Columns v4.2 Sweep. Base laser power = 40%.

		Slicing Distance (μm)					
		0.5	0.5	0.625	0.625	0.75	0.75
contour distance (μm)	0.75	X	X	X	✓	✓	✓
	0.625	X	X	X	X	✓	✓
	0.5	X	X	X	X	?	?
		75000	100000	75000	100000	75000	100000
		Scan Speed ($\mu\text{m/s}$)					

Table 4: Columns v4.3 Sweep. Base laser power = 40%.

		Slicing Distance (μm)					
		0.75	0.75	0.875	0.875	1	1
contour distance (μm)	0.75	✓	✓	✓	✓	✓	✓
	0.625	✓	✓	✓	✓	✓	✓
	0.5	✓	✓	✓	✓	✓	✓
		75000	100000	75000	100000	75000	100000
		Scan Speed ($\mu\text{m/s}$)					

Figure 3: Columns v4 Sweep. A close-up view of columns with varying levels of bubble defects is shown, with the leftmost column having more minor defects and the rightmost column missing a large volume of material.

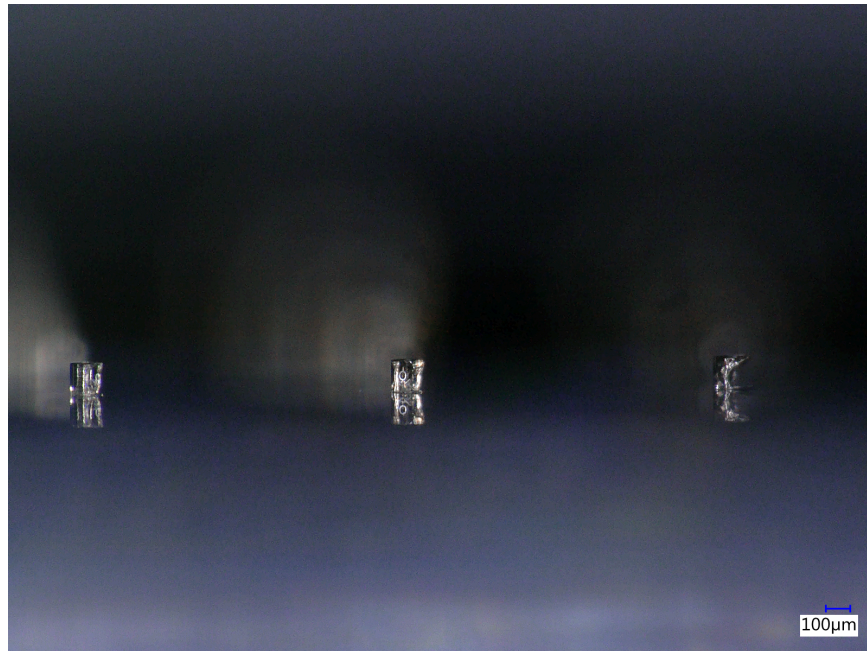


Figure 4: Columns v4 Sweep complete array. a) Bottom row corresponding to 0.25 micron contour/hatching distance (highest laser exposure), b) middle row corresponding to 0.5 micron contour/hatching distance, and c) top row corresponding to 0.75 micron contour/hatching distance (lowest laser exposure).

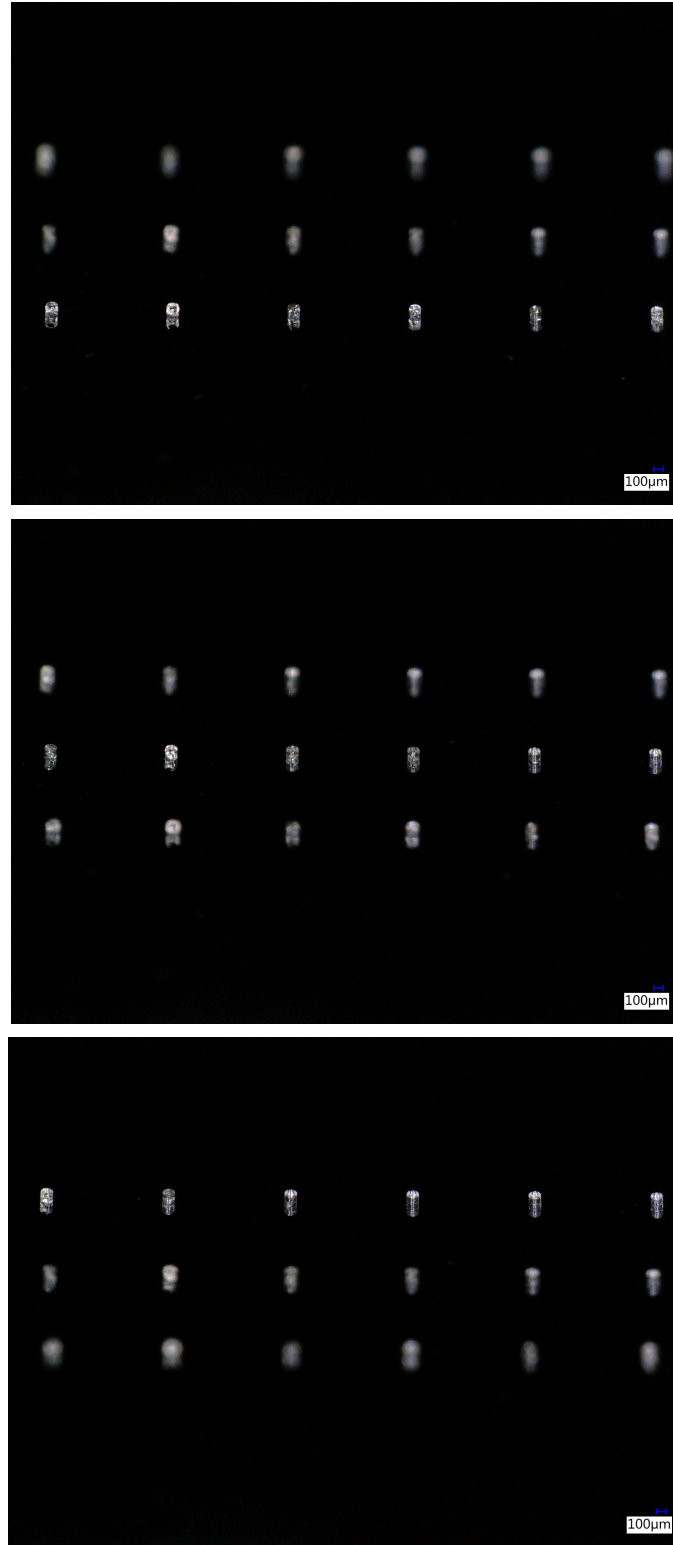
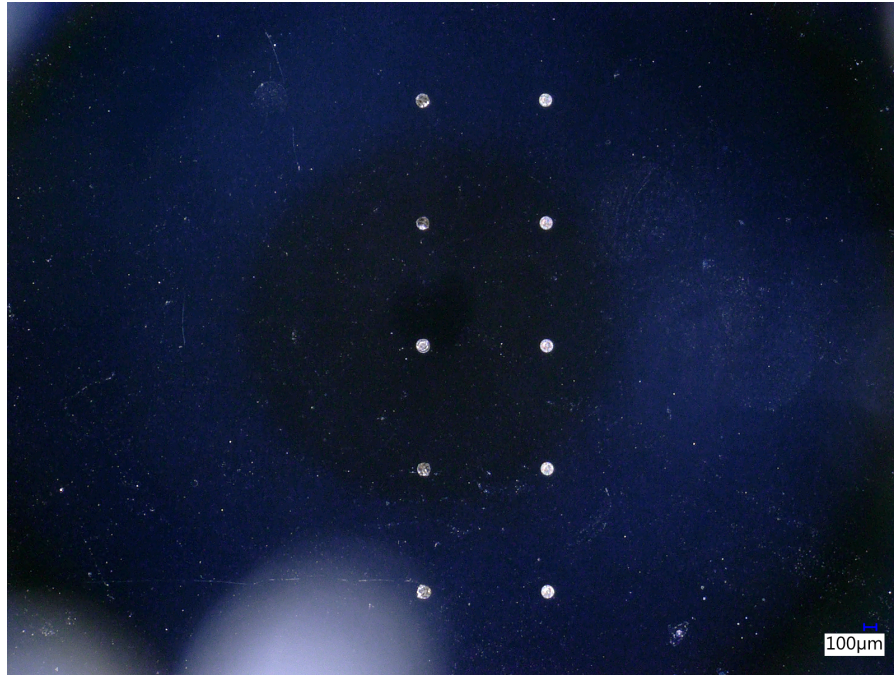


Figure 5: Columns v4.1 Power Sweep. Overhead view demonstrating the differences in the opacity of the top surfaces of the columns. The left column corresponds to higher exposure and the right column corresponds to lower exposure. Defect-free samples (right column) appear more opaque and consistent in white color.



Discussion

With the aim of serving as a user guideline analysis for Nanoscribe structure printing, one of the main takeaways is that the laser power at the base of the structure can be lowered more than previously thought (40%) with the appropriate slicing, speed, and hatching parameters without having any substrate adhesion issues during the developing process, as shown Table 2. However, it is also known that increasing laser power too much can cause major structural issues via the introduction of bubbles [15], which appear because the IP-S is being overheated and outgassing occurs. Similarly, slicing parameters appear to also have a strong effect on whether the printing generates structural bubbles or not. Therefore, it is generally advisable that the slicing parameter is chosen to be larger than the hatching distance. This follows the physical theory underpinning 2PP, as the voxel shape is elongated in the z-direction, thus reducing the slicing distance minimizes the amount of time that the photoresist is exposed by the laser. Table 4 shows that using a contour distance between 0.5 and 0.75 microns, a slicing distance between 0.75 and 1 microns, and scanning speeds between 75000 and 100000 microns/s is expected to yield a bubble-free structure.

Future Work

Due to time constraints, the samples printed have not yet undergone the elastic and plastic deformation testing with the nanoindenter, therefore this will be a spring semester endeavor for the project. Moreover, the completed nanoindentation portion of the research project has been done with the assumption that the columns have dimensions sufficiently big that they will undergo uniaxial stress, which simplifies the analysis. However, at the scales of the columns, it is worth exploring how the Young's modulus and yield stress calculation are affected by geometry using FEA in order to confidently say that the uniaxial stress is a valid assumption. Similarly, due to the nature of the capabilities and uses for the group's prints, it is useful to know how these columns with different parameters behave under shear stress, so this will also be an additional future endeavor. Finally, further exploration of the effects of development time will be conducted. The future goal is to not only provide an ample parameter list showing the varying parameters that can yield a successful print, but to quantify the degree of influence that the specific parameters have on mechanical properties such as yield strength and elastic modulus.

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